

Sanjay Dinesh

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Education

Vellore Institute of Technology, Chennai

B.Tech in Computer Science Engineering (AI & Robotics)

Aug 2023 – Present

- CGPA: 9.67 | Department Rank 3

Experience

Dreadnought Robotics, Programming Team Member

Apr 2024 – Present

- Contributed to perception, control, and systems programming for competition AUVs across **SAUVC 2025, SAUVC 2026, and TAC 2026**, covering underwater object detection, autonomous docking, and pipeline inspection tasks.
- Implemented control and path planning algorithms (**PD, A*, LSRB**) in C++/Python for real-time robotic navigation.

Microsoft Innovations Club, AIML Lead

Apr 2025 – Apr 2026

- Led a collaborative student team to build and deploy an **AI Resume Builder** (NLP-based skill extraction, formatting optimization) and **AI Academic Assistant** (LLM-based Q&A and summarization), both listed on the MIC Website.
- Promoted a supportive environment by mentoring peers in machine learning, Python, and MLOps workflows.

Autonomous Underwater Vehicles Club, President

Jul 2025 – Present

- Overseeing management events focused on teaching the industry robot development pipeline to members, bridging the gap between academic theory and practical application.

Projects

MIRA – Autonomous Underwater Vehicle | GitHub (MIRA) | GitHub (Simulator) SAUVC 2025, 2026 & TAC 2026

- Built underwater gate perception using **MobileNetV2SSD** and OpenCV; implemented **Flare Detection, Image Segmentation**, and **Autonomous ArUco Marker Docking** for SAUVC 2025 & 2026.
- Designed **YOLOv8** models for **3D Bounding Box Detection** of underwater valves, **Pipeline Detection** (simulation and real-world), and **ArUco Marker Docking** for TAC 2026.
- Validated the full programming stack using an **AUV Simulator** prior to real-world underwater deployment.

Hybrid RL-Based Cube Solver | GitHub

Completed Apr 2026

- Designed a hybrid Reinforcement Learning-based Rubik's Cube solver leveraging the **Pilgrim class** for a graph-based solving approach, achieving a rough solve time of **50–60 seconds**.
- Implemented the solution on hardware using **stepper motors** for physical cube manipulation.

4 DOF Robotic Arm for Object Classification

Completed Apr 2026

- Built a 4 DOF robotic arm that uses a **Force Sensor** to measure the force exerted on an object at the gripper during grasping, classifying it as soft or hard with an accuracy of **85%**.

Adaptive Gait Quadruped Robot

Ongoing

- Developing a quadruped robot that dynamically adjusts its gait level and stance based on real-time **IMU elevation values** to navigate uneven terrain effectively.

Skills

Languages: Python, C/C++, Java, JavaScript, HTML, CSS

Robotics & AI/ML: ROS, ROS2, Gazebo, SLAM, OpenCV, PyTorch, TensorFlow, Ollama, LangGraph, LangChain

Platforms: Docker, Git, Linux, Firebase

Soft Skills: Collaboration, Leadership, Problem-Solving, Adaptability, Communication